CSA6303

Threat Intelligence and Network Security

Assessment Tool 1: Concept Mapping (CO1 AT1)

Slot A | Questions Q5 & Q6

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats | Introduction to Threat Intelligence | Types of Threat Intelligence (Strategic, Tactical, Operational) | Threat Intelligence Lifecycle | Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

Faculty	Dr. V. Parthipan
Total Marks	50
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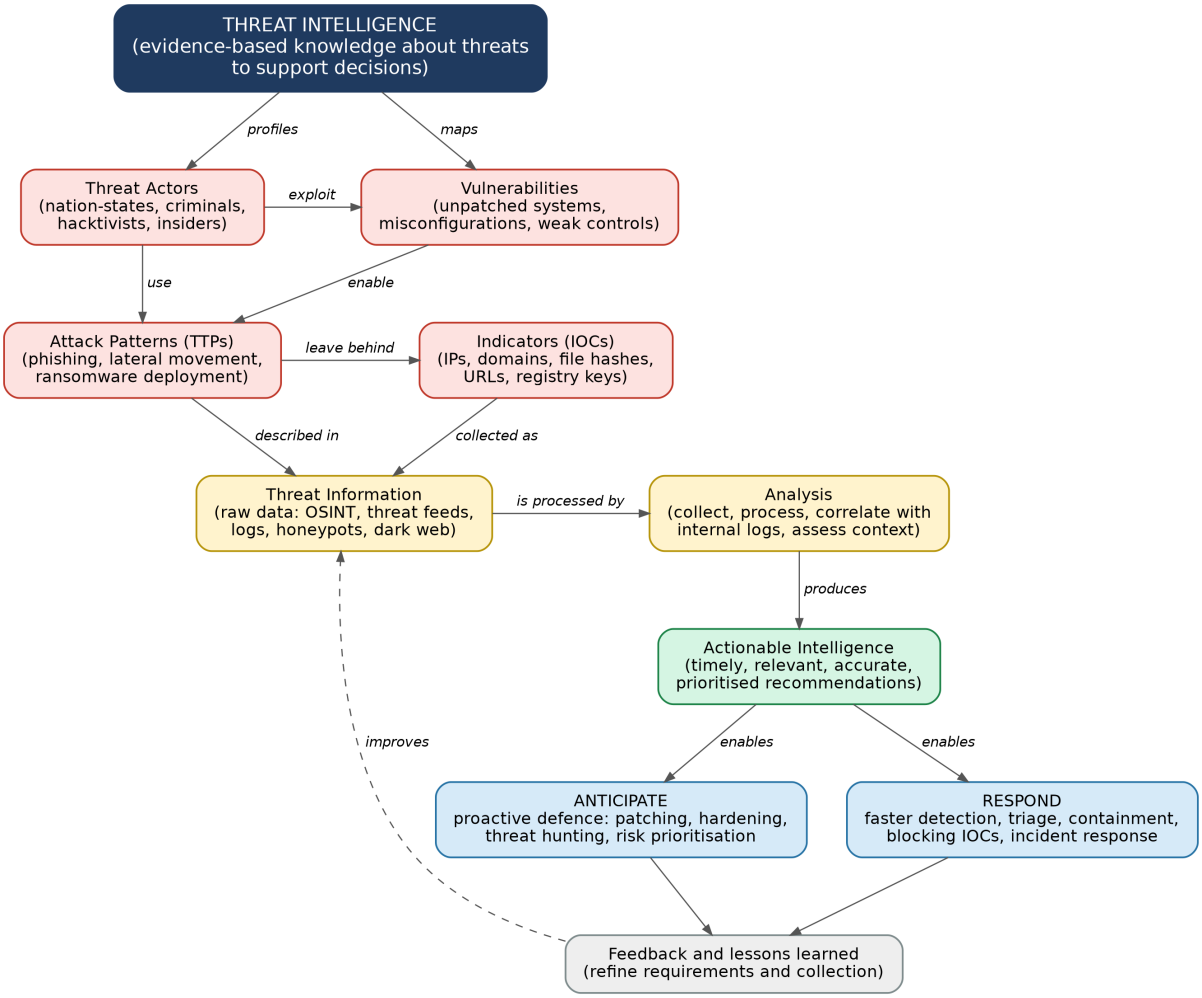
Code of Conduct

I THAMMISETTI RAJU (192465038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Q5. Concept Map: Threat Intelligence in Modern Network Security

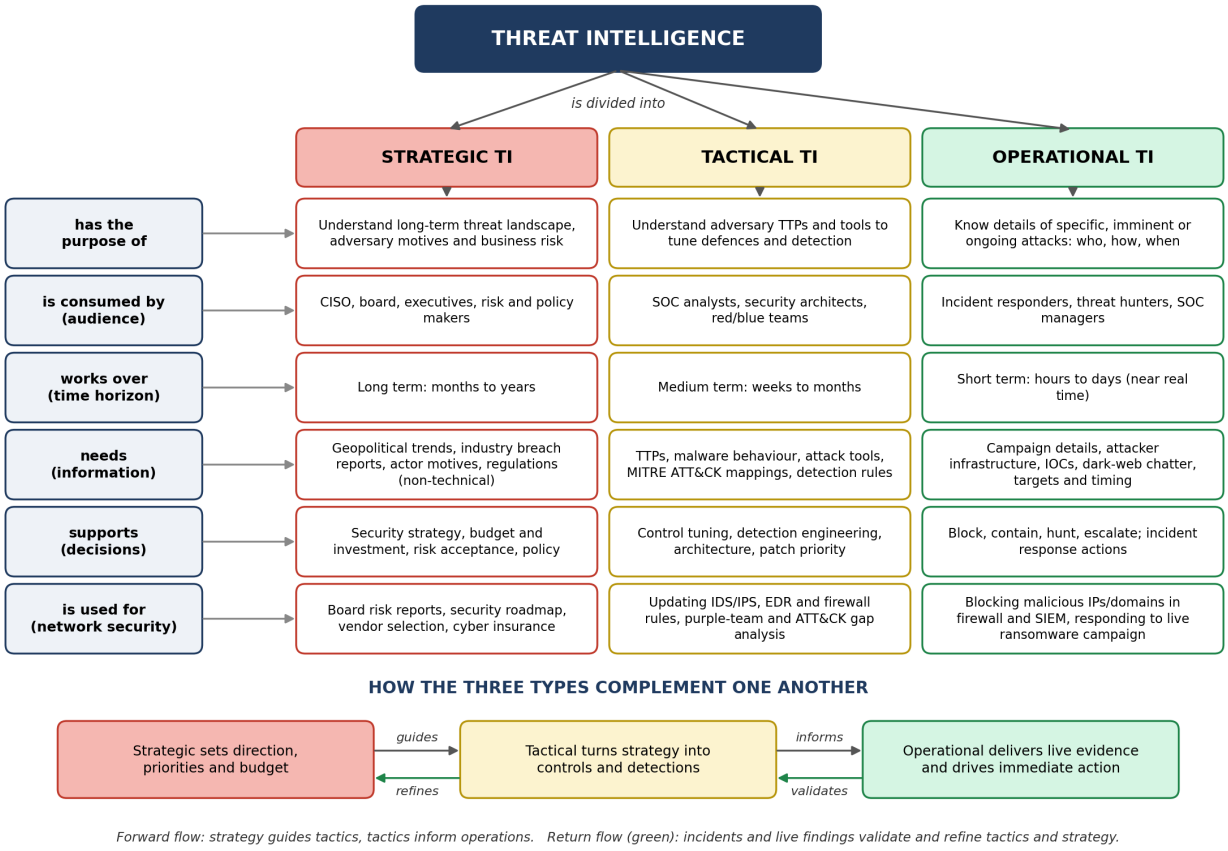
Construct a concept map explaining Threat Intelligence and its role in modern network security. Relate threat actors, indicators, vulnerabilities, attack patterns, threat information, analysis, and actionable intelligence.



Explanation. Threat actors exploit vulnerabilities and use attack patterns (TTPs), which leave behind indicators of compromise (red). These are gathered as raw threat information from feeds, OSINT and internal logs, then analysed and correlated with the organisation's own environment (yellow) to produce actionable intelligence: timely, relevant and prioritised (green). That intelligence lets the organisation **anticipate** threats through patching, hardening and threat hunting, and **respond** faster through detection, blocking and containment (blue). Lessons learned feed back into collection, so the cycle keeps improving.

Q6. Comparative Concept Map: Strategic, Tactical and Operational Threat Intelligence

Develop a comparative concept map of Strategic, Tactical, and Operational Threat Intelligence, connecting purpose, audience, time horizon, information requirements, decision-making and use cases, and analyse how the three complement one another.



Forward flow: strategy guides tactics, tactics inform operations. Return flow (green): incidents and live findings validate and refine tactics and strategy.

How they complement one another. Strategic intelligence tells leadership which threats matter to the business and where to invest. Tactical intelligence turns those priorities into concrete defences, such as detection rules and control tuning based on adversary TTPs. Operational intelligence supplies live details of active campaigns so analysts can block and contain attacks now. Findings from live incidents flow back up to validate tactics and refine strategy, so a mature security programme needs all three working together.